



Effects of milk replacer allowances and levels of starch in pelleted starter on nutrient digestibility, whole gastrointestinal tract fermentation, and pH around weaning

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ABSTRACT

The objectives of this study were to examine the effects of pelleted starter diets differing in starch and neutral detergent fiber (NDF) content when fed differing levels of milk replacer (MR) on nutrient digestibility, whole gastrointestinal tract fermentation, pH, and inflammatory markers in dairy calves around weaning. Calves were randomly assigned to 1 of 4 dietary treatments ($n = 12$ per treatment) in a 2×2 factorial design based on daily MR allowance and amount of starch in pelleted starter (SPS): 0.691 kg of MR per day [dry matter (DM) basis] with starter containing low or high starch (12.0% and 35.6% starch on DM basis, respectively), and 1.382 kg of MR per day (DM) with starter containing low or high starch. All calves were housed in individual pens with straw bedding until wk 5 when bedding was covered. Calves were fed MR twice daily (0700 and 1700 h) containing 24.5% crude protein (DM) and 19.8% fat (DM), and had access to pelleted starter (increased by 50 g/d if there were no refusals before weaning and then 200 g/d during and after weaning) and water starting on d 1. Calves arrived between 1 and 3 d of age and were enrolled into an 8-wk study, with calves undergoing step-down weaning during wk 7. Starting on d 35, an indwelling pH logger was inserted orally to monitor rumen pH until calves were dissected at the end of the study in wk 8. Higher SPS calves showed an increase in rumen pH magnitude (1.46 ± 0.07) compared with low SPS calves (1.16 ± 0.07), a decrease in rumen pH in wk 8 (high SPS: 5.37 ± 0.12 ; low SPS: 5.57 ± 0.12), and a decrease in haptoglobin in wk 8 (high SPS: 0.24 ± 0.06

g/L; low SPS: 0.49 ± 0.06 g/L). The majority of differences came from increased starter intake in general, which suggests that with completely pelleted starters the differences in starch and NDF do not elicit drastic changes in fermentation, subsequent end products, and any resulting inflammation in calves around weaning.

Key words: dairy, continuous rumen pH, metabolism, VFA, inflammatory markers

INTRODUCTION

The importance of digesta pH throughout the gastrointestinal tract (**GIT**) cannot be understated, as it regulates endogenous digestion of nutrients via enzymatic activity as well as microbial fermentation. Many factors can affect the pH of the GIT in ruminants, such as utilizing readily fermentable feedstuffs, particle size of the feed, and amount of time spent ruminating (Plaizier et al., 2008; Gressley et al., 2011; Zebeli et al., 2012). These manipulations can lead to a rapid decrease in ruminal pH, which has been connected with larger health issues such as rumenitis, laminitis, and liver abscesses, causing functional impairment of the liver in adult cattle (Nordlund et al., 1995; Lechowksi, 1997; Nocek, 1997). Interestingly, calves experiencing lower rumen pH (i.e., <5.5) are common in modern dairy calf rearing, yet they do not seem to suffer these systemic health issues (Li et al., 2012; Wood et al., 2015; Gelsinger et al., 2020; van Niekerk et al., 2021). It should be noted, though, that calves experiencing a very low rumen pH (i.e., ≤ 5.0) displayed lethargy, decreased intake, and subsequent reductions in growth (Gentile et al., 2004; Gelsinger et al., 2020).

A major contributor to lowering pH in the rumen and the lower gut (defined as the small and large intestines) is consumption of highly fermentable carbohydrates such as starch (Suarez-Mena et al., 2016;

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Gelsinger et al., 2020). Given that starch provides easily attainable ME for maintenance and growth, it should be considered a staple of diets aimed at efficient growth; however, the appropriate level to include when feeding calves remains up for debate. A meta-analysis performed by Hu et al. (2018) demonstrated a strong positive linear relationship between percent DM of starch and adjusted ADG for preweaning ($R^2 = 0.56$; $P = 0.01$) and weaned calves ($R^2 = 0.68$; $P < 0.001$). Although clear health issues may not seem apparent in calves, evidence suggests that the use of a higher-starch (42.7%) pelleted starter, compared with a lower-starch (35.3%) texturized starter, caused lower ruminal pH before, during, and after weaning; however, it should be noted that the physical form of starter, as well as the level of starch in the starter, may have influenced these results (Gelsinger et al., 2020).

When the pH of digesta is too low, the integrity of the GIT epithelial lining becomes compromised, increasing GIT permeability and allowing translocation of bacteria and lipopolysaccharides, which ultimately leads to systemic inflammation (Nocek, 1997; Plaizier et al., 2008; Khafipour et al., 2009). This reduction in GIT pH arises mainly from increased concentrate intake, which, in preweaning calves, is inversely related to milk replacer (MR) intake (Silva et al., 2019). Preweaning growth benefits of increasing MR consumption are well known (Khan et al., 2007; Sweeney et al., 2010), but it is important to consider whether calves are consuming enough CP and ME from dry feed diet to maintain growth both during and after weaning (Rauba et al., 2019). Although growth is of the utmost importance for young calves, increasing starter intake poses a risk for epithelial damage and subsequent inflammation to the rumen and lower gut in mature cows (Plaizier et al., 2008; Khafipour et al., 2009; Gressley et al., 2011) and, thus, needs to be kept in balance to maintain successful growth and health of the animal.

In light of current trends in dairy calf rearing, with producers shifting to feeding higher amounts of MR to calves, it is important to assess how this practice alters starter consumption and consequently affects nutrient digestion and GIT pH. We hypothesized that calves fed higher amounts of MR and higher-starch starter would have increased accumulations of nutrient fermentation end products, and subsequent decreases in GIT pH, as well as increased inflammatory responses during and after weaning compared with before weaning. The objectives of this study were to determine how differing amounts of MR and differing contents of starch in starters alter nutrient apparent digestibility, GIT pH around weaning, and systemic inflammatory response.

MATERIALS AND METHODS

Animals, Experimental Design, and Dietary Treatments

A total of 48 male Holstein-Friesian calves (4.4 ± 0.25 d of age) were obtained from multiple nearby dairy farms and transported to the Provimi Nurture Research Center in southwest Ohio (Lewisburg, OH) for the duration of the 8-wk study between October and December 2019. The study was performed in compliance with the guidelines set by the *Guide for the Care and Use of Agricultural Animals in Research and Teaching* (FASS, 2010), and the use of calves and experimental procedures complied with the guidelines of the Canadian Council on Animal Care (2009) and were approved by the University of Guelph Animal Care Committee (Guelph, Canada; protocol no. 4272). Details describing arrival blood sampling and housing of calves are included in Yohe et al. (2022). Upon arrival, the serum sample collected was used to measure Brix percentage via refractometry, whereupon the Brix percentage, initial BW, and location within the barn were used to stratify calves before treatment randomization using a Microsoft Excel macro-enabled randomization tool. Calves were randomly assigned to 1 of 4 dietary treatments ($n = 12$ per treatment) arranged in a 2×2 factorial based on daily MR allowance (MRA) and amount of starch in pelleted starter (SPS): (1) 0.691 kg of MR DM per day with pelleted starter containing low starch (12.0% DM basis; **LL**), (2) 0.691 kg of MR DM per day with pelleted starter containing high starch (35.6% DM basis, **LH**), (3) 1.382 kg of DM per day with pelleted starter containing low starch (**HL**), and (4) 1.382 kg of DM per day with pelleted starter containing high starch (**HH**), beginning in the second week after all calves received 0.691 kg of DM per day for the first week. All calves were fed MR (24.5% CP DM, 19.8% fat DM, 4.8 Mcal/kg of DM) at 150 g of powder per liter twice daily (0700 and 1700 h) and had access to pelleted starter (which was increased by 50 g/d if there were no refusals before weaning and then 200 g/d during and after weaning) and water starting on d 1. A more detailed description of nutrient analysis is included in Yohe et al. (2022), but, briefly, on DM basis, the low-starch starter had 20.9% CP, 44.7% NDF, 12.0% starch, and 2.9 Mcal/kg, whereas the high-starch starter had 21.5% CP, 19.7% NDF, 35.6% starch, and 3.2 Mcal/kg. All calves underwent a step-down weaning protocol consisting of only morning MR feeding starting on d 42 and ending on d 49; thus the postweaning experimental period lasted from d 49

to 57, 58, 59, or 60, depending on when calves were harvested for tissue samples.

Measurements, Sampling, and Analytical Procedures

Intake of MR and starter feed were recorded daily. Fecal samples taken for nutrient composition and apparent nutrient digestibility were composited in a subset of randomly chosen calves ($n = 6/\text{treatment}$) over a 5-d period during the preweaning (wk 6), weaning (wk 7), and postweaning (wk 8) periods. Methodology for feed and fecal sample collection, storage, and analysis is included in Yohe et al. (2022), with the addition of acid-insoluble ash (gravimetric method 920.08 in AOAC International, 2000) and sugar in starter (DuBois et al., 1956). As for the sugar (i.e., lactose) in MR it was calculated using the following equation: lactose = $100 - \text{CP} - \text{fat} - \text{ash} - 2$ (Drackley, 2008).

Separate fecal and rumen content samples were also taken weekly for VFA analysis starting in wk 4. Fecal samples were frozen and stored at -20°C until further VFA analysis. Rumen content samples were collected via oroesophageal tube at about 1200 h using an attached 60-mL syringe for aspiration. Approximately 10 mL of each rumen sample was taken after removal of any visible saliva contamination and straining rumen contents through 4 layers of cheesecloth and then immediately transferred into a sterile 15-mL polystyrene conical tube containing 25% metaphosphoric acid and stored at -20°C until further VFA and rumen microbiome analysis.

Blood samples were taken from all calves via jugular venipuncture on d 1, 14, 28, 35, 42, 49, and 56 between 0800 and 1000 h (1 to 3 h after the morning feeding) in a 10-mL glass serum tube (ref. no. 366430; BD Vacutainer). Serum samples were allowed to clot at room temperature until centrifugation at $3,000 \times g$ at 4°C for 20 min (Marathon 21000R; Fisher Scientific) and subsequently stored at -20°C until further analysis. Photometric analysis of haptoglobin was performed on a cobas 6000 series analyzer (Roche) at the University of Guelph Animal Health Laboratory (Guelph, Canada), and serum amyloid A (SAA) was analyzed using an SAA Multispecies ELISA Kit (cat. no. LSKAA0021; Invitrogen).

At the end of wk 8, calves were harvested to measure the pH of the GIT sections and to save the contents for subsequent chemical analyses. Calves were assigned to harvest on d 57, 58, 59, or 60, to balance by treatment. On the harvest day, each section of the GIT was separated, weighed full, and emptied to obtain pH, which was measured twice, and the average of the 2 measurements was used.

Indwelling Rumen pH Measurements

On d 35, all 48 calves were orally dosed with a small ruminant pH logger (SRL; Dascor Inc.) to measure rumen pH every 2 min until the end of the trial (i.e., harvest) to assess rumen pH before, during, and after weaning. The protocol was based on previous studies by Laarman et al. (2012) and Wood et al. (2015), where the SRL was calibrated before rumen insertion and then dosed while calves were held in a reclining position to allow passage into the rumen. At harvest, the SRL were retrieved and recalibrated, and the raw data were downloaded for further analysis. Average, minimum, and maximum pH were calculated on a daily basis, and further processing of the raw rumen pH data was based on the process described by Villot et al. (2018). Briefly, raw data were organized in Microsoft Excel, and a Visual Basic for Applications program was used to create a baseline pH measurement, which was calculated from the data taken before weaning (d 35 to 42); subsequent calculations were then made to compare the data obtained during weaning (d 43 to 49) and after weaning (d 50 to harvest) back to the baseline measurement to allow for assessing minutes per day that pH was 0.3 and 0.5 units below baseline, as well as area under the curve for 0.3 and 0.5 units below baseline. Data were also used to calculate minutes per day below the pH thresholds 6.0 and 5.6 as well as pH magnitude (i.e., difference between minimum and maximum pH) per day.

Statistical Analysis

Average daily gain data were used for sample size determination, as this was part of a larger study (Yohe et al., 2022), with the power analysis utilizing a power of 80%, an α -level of 0.05, and a minimum difference in BW gain of 100 g/d showing the minimal sample size of 8 calves per treatment. Absolute and relative rumen pH, GIT dissection pH, fecal composition, apparent nutrient digestibility, fecal and rumen VFA, and systemic inflammatory marker measurements were analyzed using the MIXED procedure of SAS 9.4 (SAS Institute Inc.). The study was designed as a complete randomized design with 2×2 factorial arrangement of treatments with the model including the fixed effects of MRA and SPS with repeated measures data including week (WK) and their interactions. The random effect was calf within dietary treatment, and calf was the experimental unit. All analyses included the best-fit covariance structures as determined by Akaike information criterion when appropriate; denominator degrees of freedom were not specified. Haptoglobin and SAA

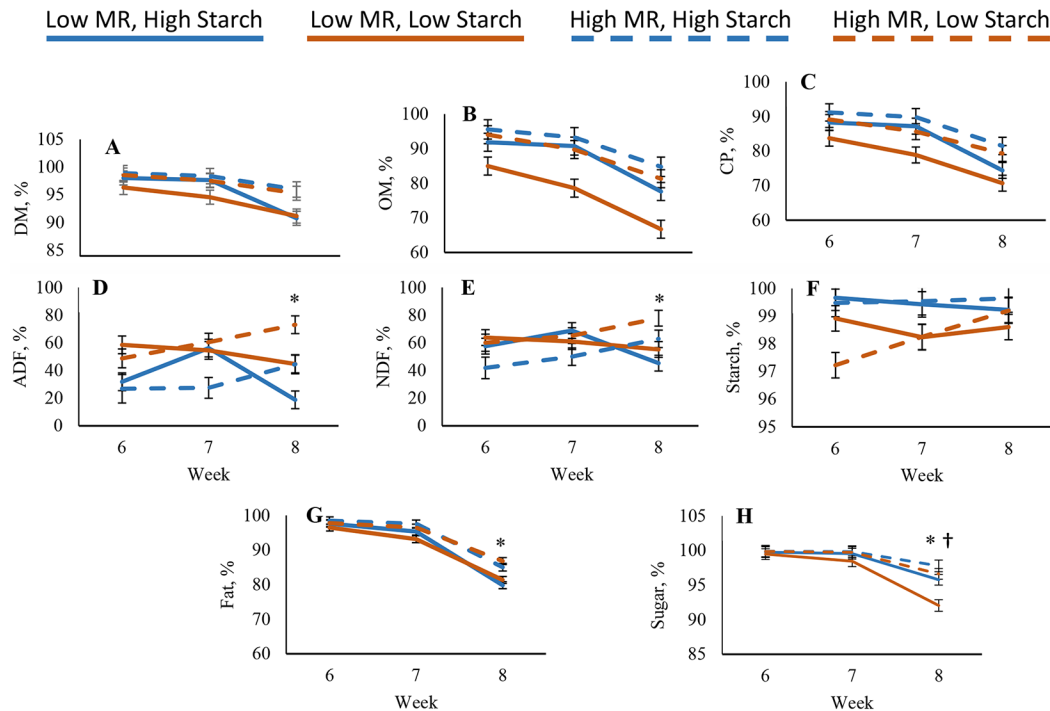


Figure 1. Weekly apparent nutrient digestibilities for (A) DM, (B) OM, (C) CP, (D) ADF, (E) NDF, (F) starch, (G) fat, and (H) sugar as a percentage of DM. Dietary treatments were as follows: low milk replacer (MR) and high-starch starter (LH); low MR and low-starch starter (LL); high MR and high-starch starter (HH); and high MR and low-starch starter (HL). LH = MR fed at 691 g of DM per day, with pelleted starter containing 35.6% starch (DM basis); LL = MR fed at 691 g of DM per day, with pelleted starter containing 12.0% starch (DM basis); HH = 1,382 g of DM per day, with pelleted starter containing 35.6% starch (DM basis); and HL = 1,382 g of DM per day, with pelleted starter containing 12.0% starch (DM basis), which started in the second week after all calves received 691 g of DM per day for the first week. All calves were fed MR containing 24.5% CP (DM) and 19.8% fat (DM) at 150 g/L twice daily (0700 and 1700 h) and had access to pelleted starter (which was increased by 50 g/d if there were no refusals before weaning and then 200 g/d during and after weaning) and water starting on d 1. Significance was declared when $P \leq 0.05$ and a tendency when $0.10 \geq P > 0.05$. Significant interactions of MR allowance $\times$ week or starch in pelleted starter $\times$ week are denoted with * or †, respectively. Error bars represent SEM.

measures included their initial values in wk 0 as a covariate. Multiple means comparisons using the PDIF option with Tukey adjustment were completed for meaningful interaction differences. Least squares means $\pm$ standard errors of the mean are reported for all variables. Pearson correlation coefficients were generated using PROC CORR in SAS, to assess the relationships between average (wk 6 to 8) rumen pH and dissection rumen measurements and between intake and pH measurements on a weekly basis. Significance was declared when $P \leq 0.05$ and a tendency when $0.10 \geq P > 0.05$.

RESULTS

Apparent Nutrient Digestibility

Weekly apparent nutrient digestibility measurements are reported in Figure 1, with samples taken before (wk 6), during (wk 7), and after weaning (wk 8). We found no differences between treatments for the interaction of

MRA $\times$ SPS. The interaction of MRA $\times$ WK saw differences in apparent digestibility for ADF ($P < 0.01$), NDF ($P < 0.01$), starch ($P = 0.03$), fat ($P = 0.01$), and sugar ($P = 0.02$), with a difference in fat ($P = 0.03$) and a tendency for sugar ($P = 0.08$) apparent digestibility for the interaction of SPS $\times$ WK. For the main effect of MRA, we observed increases in apparent digestibility for DM ($P = 0.01$), OM ($P < 0.01$), CP ($P < 0.01$), fat ($P < 0.01$), and sugar ($P = 0.02$) in high-MRA calves. As for the main effect of SPS, increases occurred in apparent ADF ($P < 0.01$) and NDF digestibility ($P = 0.05$) for low-SPS calves, with increases in OM ($P = 0.01$), CP ($P = 0.05$), and starch ($P = 0.01$) and a tendency in sugar digestibility ($P = 0.07$) in high-SPS calves.

Fecal Composition

Weekly fecal nutrient compositions are reported in Supplemental Figure S1 (<https://doi.org/10.6084/m9>

[.figshare.19192769.v1](#)). For the interaction of MRA $\times$ SPS we found a difference for ADF ($P = 0.04$) and a tendency for NDF ($P = 0.07$) in LL calves to be higher than the other treatments and a tendency for starch ($P = 0.10$) to be higher in the LH calves. The interaction of MRA $\times$ WK saw differences in fecal OM ($P = 0.01$), CP ($P < 0.01$), ADF ($P = 0.01$), NDF ($P < 0.01$), fat ($P < 0.01$), and ash ($P = 0.01$), but no SPS $\times$ WK interactions were detectable. These MRA $\times$ WK interactions are shown in Supplemental Figure S1, but after performing the post-hoc Tukey's test, we found no differences in OM or ash between MRA levels. As for the main effect of MRA increases occurred in CP ($P < 0.01$) and fat ($P < 0.01$) in high-MRA calves, along with an increase in ADF ($P = 0.01$) and NDF ($P < 0.01$) in low-MRA calves. For the main effect of SPS, increases occurred in OM ($P < 0.01$), ADF ($P < 0.01$), and NDF ($P < 0.01$) in low-SPS calves, as well as increases in CP ($P < 0.01$), starch ($P = 0.05$), fat ($P < 0.01$), and ash ($P < 0.01$) in high-SPS calves.

Rumen and Fecal VFA

Weekly rumen and fecal samples that were taken and analyzed for total VFA and proportions of acetate, propionate, and butyrate are shown in Figure 2. For rumen measurements, the MRA $\times$ WK interaction exhibited differences for total VFA, proportions of acetate and propionate, and a tendency for proportion of butyrate. The main effect of MRA showed increases in total VFA, proportion of propionate, and a tendency for proportion of butyrate in low-MRA calves, and an increase in proportion of acetate in high-MRA calves, whereas for the main effect of SPS we found an increase in proportion of acetate for low-SPS calves and an increase in proportion of propionate for high-SPS calves.

For fecal measurements, the MRA $\times$ WK interaction exhibited a difference for proportion of acetate and a tendency for proportion of butyrate. However, upon performing the multiple means comparisons we found no differences in fecal acetate concentrations between MRA levels within week. The interaction of MRA $\times$ SPS showed an increase in proportion of acetate and a decrease in proportion of propionate for LL calves compared with the other treatments. The main effect of MRA showed an increase in proportion of acetate in low-MRA calves and a tendency for an increase in proportion of butyrate in high-MRA calves, whereas for the main effect of SPS, we detected an increase in proportion of acetate in low-SPS calves and increases in proportions of propionate and butyrate in high-SPS calves.

Gastrointestinal Tract pH

Rumen pH was measured continuously every 2 min from d 35 until harvest (d 57 to 60), with multiple pH variables assessed during this period (Table 1). No differences were detectable for the interactions or main effects of MRA and SPS for absolute measurements of minimum, mean, and maximum pH or time that pH was below 5.6, although all measurements saw differences by WK. We detected an interaction of MRA $\times$ WK for time pH was below 6.0, although after performing multiple means comparisons no differences were detected within WK. We found an interaction of SPS $\times$ WK for magnitude, with an increase in HH compared with HL calves during wk 7 (i.e., during weaning), as well as the main effect of SPS, which saw an increase in high-SPS compared with low-SPS calves (1.47 ± 0.11 and 1.16 ± 0.11 , respectively) throughout wk 6 to 8. Time below pH 6.0 and pH magnitude also showed WK effects similar to the other absolute pH measurements. For the relative pH measurements (i.e., min/d and area under the curve either 0.3 or 0.5 units below baseline), an effect of WK was detected for every measurement but no other interaction or main effects of MRA or SPS. We observed a tendency for the measurement of time 0.5 units below baseline (min/d) for the interaction of MRA $\times$ WK, but after performing the multiple means comparisons no tendencies were noted.

The pH of the entire GIT determined at harvest is reported in Table 2. For the limited differences observed, we found an effect of MRA on reticulorumen pH, with low-MRA calves having lower pH, as well as a tendency for proximal jejunum, with low MRA having lower pH as well. We note a few more differences for the effect of SPS, with low-SPS calves having decreased abomasal pH, whereas high-SPS calves had increased cecal pH. Tendencies included high-SPS calves having lower reticulorumen and colonic pH.

Inflammatory Markers

Weekly blood samples were taken to measure the inflammatory markers haptoglobin and SAA (Figure 3). For haptoglobin, we observed an effect of MRA, with high-MRA calves having an increased level of haptoglobin throughout the experiment (high MRA = 0.24 ± 0.02 g/L, low MRA = 0.17 ± 0.02 g/L), whereas no main effect of SPS occurred. An interaction effect of SPS $\times$ WK was detectable for higher haptoglobin in low-SPS compared with high-SPS calves in wk 8. For SAA, no differences occurred in the main effects of MRA or SPS or the interaction of MRA $\times$ WK or SPS $\times$ WK. We detected a tendency for a difference in

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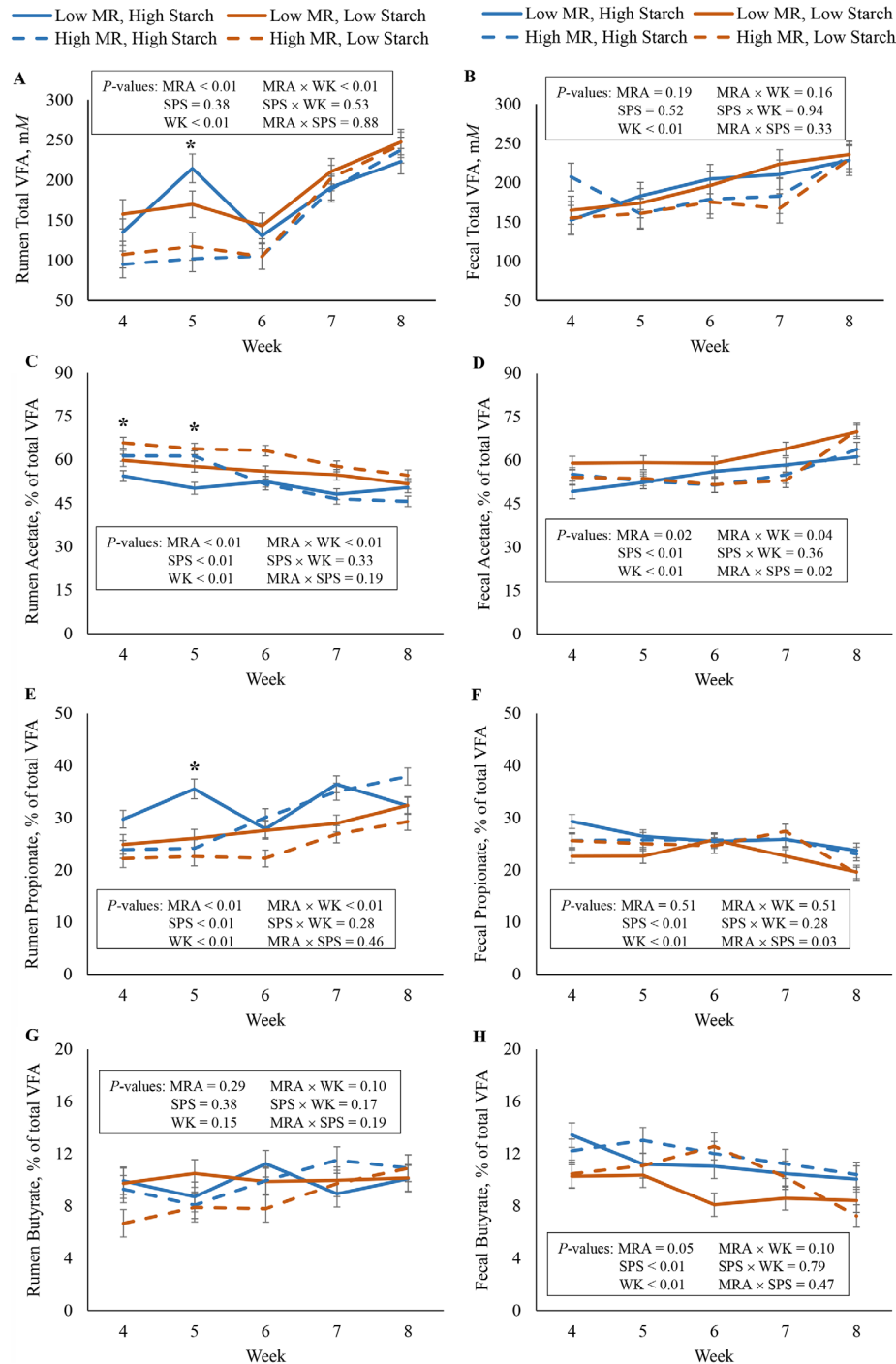


Figure 2. Weekly rumen and fecal total VFA concentrations and proportions for (A) rumen total VFA, (B) fecal total VFA, (C) percent rumen acetate, (D) percent fecal acetate, (E) percent rumen propionate, (F) percent fecal propionate, (G) percent rumen butyrate, and (H) percent fecal butyrate. Dietary treatments were as follows: low milk replacer (MR) and high-starch starter (LH), low MR and low-starch starter (LL), high MR and high-starch starter (HH), and high MR and low-starch starter (HL). LH = MR fed at 691 g of DM per day, with pelleted starter containing 35.6% starch (DM basis); LL = MR fed at 691 g of DM per day, with pelleted starter containing 12.0% starch (DM basis); HH = 1,382 g of DM per day, with pelleted starter containing 35.6% starch (DM basis); and HL = 1,382 g of DM per day, with pelleted starter containing 12.0% starch (DM basis), which started in the second week after all calves received 691 g of DM per day for the first week. All calves were fed MR containing 24.5% CP (DM) and 19.8% fat (DM) at 150 g/L twice daily (0700 and 1700 h) and had access to pelleted starter (which was increased by 50 g/d if there were no refusals before weaning and then 200 g/d during and after weaning) and water starting on d 1. MRA = MR allowance; SPS = starch amount in pelleted starter; WK = week. Significance was declared when $P \leq 0.05$ and a tendency when $0.10 \geq P > 0.05$. * denotes MRA $\times$ WK effect. Any MRA $\times$ WK difference not noted is due to lack of difference after multiple means comparison performed. Error bars represent SEM.

Table 1. Minimum, mean, maximum, time under specific pH_i, and time and area under the curve below baseline around weaning for calves fed different amounts of milk replacer (MR) or starch concentrations in starter during the 8-wk study

pH measurement	Dietary treatment ¹						Test of fixed effects, <i>P</i> -value ²					
	Low MR			High MR								
	High starch	Low starch	High starch	Low starch	SEM	MRA ³	SPS ⁴	MRA × SPS	WK ⁵	MRA × WK	SPS × WK	
Absolute ⁶												
Minimum												
Wk 6 ^a	5.23	5.21	5.26	5.42	0.11	0.66	0.41	0.58	<0.01	0.34	0.62	
Wk 7 ^b	5.17	5.20	5.09	5.33	0.13							
Wk 8 ^c	5.04	5.13	5.05	5.16	0.16							
Wk 6–8	5.15	5.19	5.15	5.29	0.09							
Mean												
Wk 6 ^a	5.86	5.81	6.05	5.99	0.11	0.33	0.65	0.77	<0.01	0.15	0.98	
Wk 7 ^a	5.92	5.83	5.96	5.93	0.13							
Wk 8 ^b	5.73	5.58	5.76	5.80	0.17							
Wk 6–8	5.84	5.74	5.91	5.90	0.12							
Maximum												
Wk 6 ^b	6.56	6.50	6.74	6.57	0.12	0.44	0.12	0.99	<0.01	0.82	0.20	
Wk 7 ^a	6.73	6.55	6.89	6.58	0.15							
Wk 8 ^b	6.71	6.23	6.73	6.49	0.21							
Wk 6–8	6.68	6.41	6.75	6.54	0.13							
Magnitude ⁷												
Wk 6 ^b	1.31 _{AB}	1.21	1.47	1.07	0.11	0.35	<0.01	0.34	<0.01	0.32	0.03	
Wk 7 ^a	1.47 _{AB}	1.22 ^{AB}	1.75 ^A	1.16 ^B	0.11							
Wk 8 ^b	1.33	1.04	1.48	1.23	0.11							
Wk 6–8	1.37	1.16	1.57	1.15	0.11							
Time pH <5.6 (min/d)												
Wk 6 ^c	443	436	250	202	88	0.13	0.61	0.30	<0.01	0.34	0.92	
Wk 7 ^b	496	557	488	333	103							
Wk 8 ^a	717	812	783	556	112							
Wk 6–8	543	591	493	354	92							
Time pH <6.0 (min/d)												
Wk 6 ^c	883	892	633	697	98	0.17	0.52	0.55	<0.01	0.03	0.66	
Wk 7 ^b	839	982	854	893	101							
Wk 8 ^a	1,041	1,231	1,062	975	101							
Wk 6–8	915	1,025	839	848	91							
Relative												
0.3 bb ⁸ (min/d)												
Wk 6 ^c	257	282	306	171	37	0.27	0.43	0.11	<0.01	0.11	0.90	
Wk 7 ^b	319	431	596	391	101							
Wk 8 ^a	537	599	860	629	120							
Wk 6–8	363	430	574	386	79							
0.5 bb (min/d)												
Wk 6 ^c	122	95	153	62	31	0.16	0.18	0.46	<0.01	0.08	0.67	
Wk 7 ^b	220	190	386	212	93							
Wk 8 ^a	350	263	646	427	120							
Wk 6–8	225	178	383	224	77							
0.3 bb (AUC ⁹)												
Wk 6 ^c	67	62	89	28	19	0.60	0.14	0.69	<0.01	0.67	0.41	
Wk 7 ^b	147	106	223	103	62							

Continued

Table 1 (Continued). Minimum, mean, maximum, time under specific pH, and time and area under the curve below baseline around weaning for calves fed different amounts of milk replacer (MR) or starch concentrations in starter during the 8-wk study

pH measurement	Dietary treatment ¹						Test of fixed effects, <i>P</i> -value ²							
	Low MR			High MR			SEM	MRA ³	SPS ⁴	MRA × SPS	WK ⁵	MRA × WK	SPS × WK	
	High starch	Low starch		High starch	Low starch									
Wk 8 ^a	300	155		365	215		99							
Wk 6–8	165	105		219	110		60							
0.5 bb (AUC)														
Wk 6 ^c	30	28		43	6		14	0.83	0.13	0.92	<0.01	0.86		0.34
Wk 7 ^b	94	47		124	44		44							
Wk 8 ^a	212	69		212	109		79							
Wk 6–8	107	47		122	50		47							

^{a-c}Lowercase superscripts within a column represent difference between weeks, regardless of treatment.^{A,B}Uppercase superscripts within a row represent difference between interaction of MRA × WK or SPS × WK.¹Dietary treatments were as follows: low MR and high-starch starter (LH), low MR and low-starch starter (LL), high MR and high-starch starter (HH), and high MR and low-starch starter (HL). LH = MR fed at 691 g of DM per day, with pelleted starter containing 35.6% starch (DM basis); LL = MR fed at 691 g of DM per day, with pelleted starter containing 12.0% starch (DM basis); HH = 1,382 g of DM per day, with pelleted starter containing 35.6% starch (DM basis); and HL = 1,382 g of DM per day, with pelleted starter containing 12.0% starch (DM basis), which started in the second week after all calves received 691 g of DM per day for the first week. All calves were fed MR containing 24.5% CP (DM) and 19.8% fat (DM) at 150 g/L twice daily (0700 and 1700 h) and had access to pelleted starter (which was increased by 50 g/d if there were no refusals before weaning and then 200 g/d during and after weaning) and water starting on d 1.²Significance was declared when $P \leq 0.05$ and a tendency when $0.10 \geq P > 0.05$.³MRA = MR allowance.⁴SPS = amount of starch in the pelleted starter.⁵WK = week.⁶pH minimum, mean, maximum, and magnitude values were all calculated on a daily basis.⁷Magnitude = difference between pH maximum and pH minimum on a daily basis.⁸bb = below baseline.⁹AUC = area under the curve.

Table 2. Dissection pH data of calves at 8 wk fed different amounts of milk replacer (MR) or starch concentrations in starter

Gastrointestinal region	Dietary treatment ¹				SEM	Test of fixed effects, <i>P</i> -value ²		
	Low MR		High MR			MRA ³	SPS ⁴	MRA × SPS
	High starch	Low starch	High starch	Low starch				
Reticulorumen	5.14	5.50	5.60	5.63	0.12	0.02	0.10	0.17
Omasum	5.39	5.60	5.55	5.45	0.18	0.97	0.75	0.37
Abomasum	3.66	2.87	3.48	3.22	0.20	0.67	0.01	0.19
Duodenum	5.58	5.73	5.73	5.55	0.19	0.95	0.93	0.36
Proximal jejunum	5.93	6.04	6.19	6.06	0.08	0.07	0.94	0.11
Distal jejunum	7.46	7.49	7.43	7.53	0.07	0.96	0.35	0.63
Ileum	7.54	7.59	7.44	7.48	0.07	0.19	0.55	0.94
Cecum	6.33	6.60	6.38	6.67	0.10	0.55	0.01	0.92
Colon	6.39	6.62	6.54	6.68	0.10	0.33	0.07	0.67

¹Dietary treatments were as follows: low MR and high-starch starter (LH), low MR and low-starch starter (LL), high MR and high-starch starter (HH), and high MR and low-starch starter (HL). LH = MR fed at 691 g of DM per day, with pelleted starter containing 35.6% starch (DM basis); LL = MR fed at 691 g of DM per day, with pelleted starter containing 12.0% starch (DM basis); HH = 1,382 g of DM per day, with pelleted starter containing 35.6% starch (DM basis); and HL = 1,382 g of DM per day, with pelleted starter containing 12.0% starch (DM basis), which started in the second week after all calves received 691 g of DM per day for the first week. All calves were fed MR containing 24.5% CP (DM) and 19.8% fat (DM) at 150 g/L twice daily (0700 and 1700 h) and had access to pelleted starter (which was increased by 50 g/d if there were no refusals before weaning and then 200 g/d during and after weaning) and water starting on d 1.

²Significance was declared when $P \leq 0.05$ and a tendency when $0.10 \geq P > 0.05$.

³MRA = MR allowance.

⁴SPS = amount of starch in the pelleted starter.

MRA × SPS, but after performing the multiple means comparisons there was no tendency to report.

Correlation Results

Pearson correlation coefficients were generated and are presented in Supplemental Tables S1 through S7 (<https://doi.org/10.6084/m9.figshare.19192769.v1>). Many relationships were observed between total rumen pH measurements throughout wk 6 to 8 and rumen measurements or rumen samples taken at dissection, with notable observations including the following: rumen 2-dimensional papillae area having a weak positive relationship to minimum ruminal pH, and full reticulorumen weight and reticulorumen as a percentage of BW having a strong negative correlation with rumen pH magnitude (Supplemental Table S1). For weekly rumen pH measurements, many strong and weak negative relationships occurred between many pH metrics and ruminal VFA measurements, with a pattern of these relationships weakening as calves grew older (Supplemental Tables S2, S3, and S4). The subset of calves that were used for apparent nutrient digestibility and fecal composition measures were used to correlate these results with weekly rumen pH measurements, which are shown in Supplemental Tables S5, S6, and S7. Many positive and negative relationships are shown in the supplemental tables; notable observations include DM digestibility having a strong negative relationship with minimum and mean pH in wk 6 (before weaning), a strong negative relationship between minimum rumen

pH and fecal starch in wk 7 (during weaning), and a strong negative relationship between rumen pH magnitude and fecal NDF in wk 8 (after weaning).

DISCUSSION

The goal of this study was to examine whether feeding a higher amount of MR and a higher-starch starter would lead to an increased inflammatory response due to a buildup of nutrient fermentation end products and subsequent decrease in GIT pH during and after weaning. These data are from a larger study that has previously reported intake, growth, and visceral tissue growth and development from the same calves (Yohe et al., 2022). Our findings indicate limited effects at the individual treatment levels but multiple differences in nutrient fermentation and subsequent pH and systemic inflammation at the main effects and interactions of MRA, SPS, and WK.

One of the most important aspects for proper rumen function is pH, which was measured using an indwelling bolus around the time of weaning. A notable difference measured in rumen pH (other than the main effect of WK) was for the interaction of SPS × WK and the main effect of SPS for pH magnitude, which was increased in HH compared with HL calves during weaning and increased in high-SPS compared with low-SPS calves, respectively. These findings suggest 2 things: (1) on a high level of MR, calves fed a high-starch pelleted starter experience larger pH swings during weaning than calves fed low-starch pelleted starter, and (2)

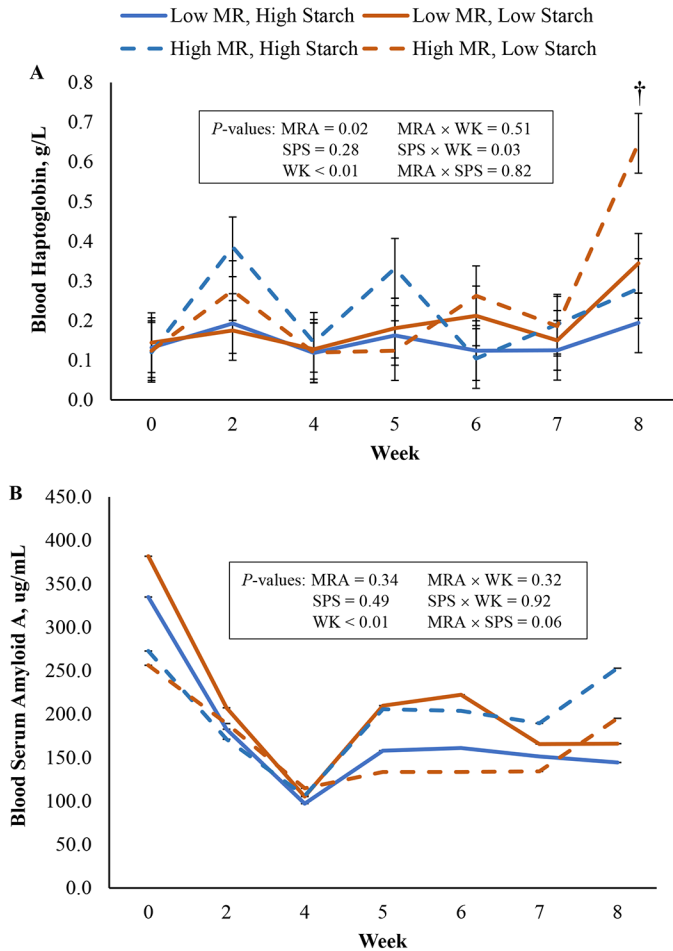


Figure 3. Weekly blood measurements for (A) haptoglobin and (B) serum amyloid A. Dietary treatments were as follows: low milk replacer (MR) and high-starch starter (LH), low MR and low-starch starter (LL), high MR and high-starch starter (HH), and high MR and low-starch starter (HL). LH = MR fed at 691 g of DM per day, with pelleted starter containing 35.6% starch (DM basis); LL = MR fed at 691 g of DM per day, with pelleted starter containing 12.0% starch (DM basis); HH = 1,382 g of DM per day, with pelleted starter containing 35.6% starch (DM basis); and HL = 1,382 g of DM per day, with pelleted starter containing 12.0% starch (DM basis), which started in the second week after all calves received 691 g of DM per day for the first week. All calves were fed MR containing 24.5% CP (DM) and 19.8% fat (DM) at 150 g/L twice daily (0700 and 1700 h) and had access to pelleted starter (which was increased by 50 g/d if there were no refusals before weaning and then 200 g/d during and after weaning) and water starting on d 1. MRA = MR allowance; SPS = starch amount in pelleted starter; WK = week. Significance was declared when $P \leq 0.05$ and a tendency when $0.10 \geq P > 0.05$. † denotes an SPS × WK effect. Error bars represent SEM.

overall, calves fed a high-starch pelleted starter show the largest pH swings during weaning, when starter consumption increases dramatically. Gelsinger et al. (2020) fed starters differing in starch and form and observed that the pelleted, higher-starch starter caused lower overall mean pH (5.37) compared with the texturized, lower-starch starter (5.63), as well as a decrease in

pH in the pelleted starter-fed calves in the hours after feeding. The higher starch in our study did not seem to elicit the lower rumen pH values seen in adult cows around parturition (Penner et al., 2007, 2011), but this is most likely explained by DM disappearance rates. Laarman et al. (2012) and Gelsinger et al. (2019) both demonstrated that in situ DM disappearance was an important factor for determining rumen pH in calves.

With regard to intake, it is interesting to note that, before weaning, all measures of intake (i.e., DMI and DM intakes of NDF, starch, ME, and CP) tended to have a weak negative relationship with mean pH and a weak positive relationship with minutes that rumen pH was below 6.0 and 5.6 (except for a lack of relationship between NDF intake and minutes <5.6). During and after weaning, however, these relationships were no longer present. Similarly, before weaning, many of the rumen pH measures had at least a weak negative relationship with all rumen VFA measures (i.e., total VFA, acetate, propionate, and butyrate), but these relationships lessened during and after weaning (number of negative relationships between rumen pH measures and rumen VFA measures that were at least tendencies, by week: wk 6 = 17, wk 7 = 11, wk 8 = 7). These findings suggest that a more developed rumen environment (i.e., microbial community and rumen tissue) may be necessary to adequately absorb (Williams et al., 1987) and buffer (Yang and Beauchemin, 2006) the rumen in response to the increase in microbial fermentation end products, thus allowing rumen pH to increase to a level that is more typically observed in adult ruminants (as reviewed in Baldwin VI et al., 2004; van Niekerk et al., 2021).

The pH of different GIT segments taken at dissection in wk 8 showed a lower rumen pH in low-MRA calves compared with high-MRA calves, which is a well-established concept (Li et al., 2012; Gelsinger et al., 2020). Also, high-SPS calves tended to have decreased rumen pH compared with low-SPS calves, which has been shown recently by Gelsinger et al. (2020) and confirmed here using pelleted starters for both treatments. It is important to note that these differences in rumen pH in the current study were from samples taken at an individual time point, whereas, to address the large variability observed in rumen pH, samples should be taken throughout the day (Wood et al., 2015; Khan et al., 2016; Gelsinger et al., 2020; van Niekerk et al., 2021).

A main factor determining pH throughout the GIT is how well nutrients are digested to be available for subsequent absorption and metabolism. High-MRA calves had increased apparent digestibility for DM, OM, CP, fat, and sugar, which all stem from increased digestibility in the lower gut of MR ingredients compared

with starter ingredients in the developing rumen (Davis and Drackley, 1998). Interestingly, ADF and NDF digestibility were higher in high-MRA calves in wk 8, which is contrary to other studies that demonstrated decreased ADF and NDF digestibility after weaning in calves fed higher levels of MR (Dennis et al., 2018; Hu et al., 2020; van Niekerk et al., 2020). Hu et al. (2020), however, observed that by 16 wk (approx. 8 wk after weaning) the digestibility of ADF and NDF were the same regardless of prior MR feeding rate. We observed increased digestibility of DM, OM, CP, and fat in high-MRA calves, which was similar to Quigley et al. (2018) in the preweaning period (wk 5) but not postweaning (wk 8), and contrary to what van Niekerk et al. (2020) observed during wk 5 preweaning and wk 8 postweaning (except wk 5 OM digestibility), despite feeding similar MR diets with either a pelleted or a texturized starter. The differing intakes of MR or starter ingredients provide likely explanations for any differences between previous studies, although the much lower starter intakes for both treatments in our study might potentially provide another explanation. For example, our low- and high-MRA calves consumed starter at 0.48 ± 0.07 and 0.21 ± 0.07 kg/d, respectively, which were much lower compared with a meta-analysis utilizing studies where calves were fed similar MR rates (comparable to low MRA: 0.70 ± 0.02 kg/d, and comparable to high MRA: 0.47 ± 0.02 kg/d; Hu et al., 2020).

In relation to NFC intake, the digestibility of ADF and NDF were increased in low-SPS calves, whereas starch digestibility, and a tendency for sugar digestibility, was increased in high-SPS calves, which is most likely attributable to established microbial communities that are better able to break down fiber (van Niekerk et al., 2020). Quigley et al. (2018) provided further support that intake of both fiber (NDF and ADF) and starch are related to their digestibility by demonstrating that cumulative intake of both ingredients is the best indicator of their digestion.

Measuring differences in nutrient fermentation was another goal of this study, with fecal composition results showcasing some of these differences. It should be noted that these fecal results were not from total collections, so excretion of nutrients must be interpreted as a proportion, and not the total amount, within the sampling time periods. These differences included the MRA $\times$ SPS interaction, with LL calves having higher fecal ADF (and tendency for higher fecal NDF), whereas LH calves had a tendency for increased fecal starch, both of which are due to the increased ADF and NDF and starch consumed in their starter, respectively. Increased fecal CP and fat resulted from their higher MR intake before and during weaning (wk 6 and 7), whereas the increase in fecal ADF and NDF in

low-MRA calves was from their increased starter intake during this same period. Hill et al. (2008) assessed diets containing a lower amount of MR fed (441 g of powder DM per day at 20% CP and 21% fat) to higher amount of MR fed (and higher CP and fat; 951 or 1,431 g of powder DM per day at either 28% CP and 20% fat or 27% CP and 28% fat) and found no difference in fecal N output between treatments, although the calves fed lower MR consumed much less N compared with the calves fed higher MR. Although our high-MRA calves had a higher proportion of CP in their feces than the low-MRA calves, it is worth noting that total excretion was not measured in our study, which may have shown similar results to those of Hill et al. (2008).

As for starch levels in the starter, the increase in fecal OM, ADF, and NDF in low-SPS calves makes sense based on the increased ADF and NDF in the starter. This was also shown by Gelsinger et al. (2019), where fecal NDF output increased with higher NDF intake. The increase in fecal starch content in high-SPS calves appears to have been a consequence of increased starch intake, as in Gelsinger et al. (2019). Interestingly, Dennis et al. (2017) observed a positive relationship between fecal starch and starch digestibility preweaning, but an inverse relationship between starch intake and fecal starch postweaning, whereas fecal starch and digestibility were both inversely correlated preweaning and postweaning in the current study. This suggests that other factors (e.g., MR intake and passage rate) most likely played a role in the fecal starch difference between starch levels fed to calves in the current study.

Although we did not study microbial composition, we assessed the fermentation end products they create, VFA, which were found to be affected by both MRA and SPS treatments. The general trend throughout the study was that an increase in starter consumption led to an increase in total rumen VFA, and proportions of propionate and butyrate as well, but proportion of acetate decreased. This relationship between starter intake and both propionate and butyrate compared with NDF or forage intake and acetate has been observed in many previous calf studies (as reviewed in Xiao et al., 2020). In the current study, increased proportion of acetate was observed in low-SPS calves (i.e., increased NDF intake), which was also observed by Suárez et al. (2006), with increased ruminal acetate resulting from starter containing higher NDF compared with starch. It is logical that total VFA increased in all calves throughout the study due to increased starter consumption (Tamate et al., 1962; Stobo et al., 1966). In the current study, a diminishing pattern between rumen VFA measures and intake over time was observed, where DM, NDF, starch, ME, and CP intake had at least a weak positive relationship 90% of the time in wk

6, whereas in wk 7, at least a weak positive relationship was observed only 65% of the time, with 1 being a weak negative relationship, and in wk 8, positive relationships occurred only 25% of the time. These results suggest that early on in the developing rumen, when the calf is just beginning to increase starter consumption, the rumen environment is more susceptible to change compared with when more substrate is present in the rumen and a more stable microbial community provides a more consistent supply of VFA (as reviewed in Malmuthuge et al., 2015).

Assessing fecal VFA shows hindgut fermentation patterns, with the results suggesting that the increased proportion of acetate and the decreased proportion of propionate in LL calves may have been the result of NDF reaching the hindgut and allowing for more acetate production (as demonstrated in the rumen by Suárez et al., 2006, and in our rumen VFA results for low-SPS calves), as these calves were consuming more starter containing NDF before and during weaning. Increased proportions of propionate and butyrate in high-SPS calves is most likely due to increased starch reaching the hindgut (as evidenced by higher fecal starch in high-SPS calves) and was also observed by Ren et al. (2020) in diets containing increased rumen-escape starch. For all fecal VFA it would be interesting to examine the differences in microbial populations present, which may have showcased differing levels of amylolytic and fibrolytic bacteria in the hindgut (Ren et al., 2020; van Niekerk et al., 2020) when considering the differences in starch and NDF digestibility between low- and high-SPS calves.

When discussing microbial fermentation, consideration needs to be given to any inflammation resulting from excessive VFA and other fermentation end products that are created, as well as the potential impairment in gut barrier function that may catalyze systemic health issues. Although the overall health of the calves was not negatively affected by the treatments in the study, low-SPS calves had the highest haptoglobin measures in wk 8, which may suggest an acute inflammatory response during this week. Although this does not explain why low-SPS calves specifically had increased haptoglobin in wk 8, all calves showed an increase in haptoglobin from wk 7 to 8 (also drastically increasing their starter intake during this time), but all measures of intake were negatively correlated with haptoglobin during wk 8. Also, contrary to results from Quigley et al. (2006) and Obeidat et al. (2013), we observed an increase in haptoglobin for high-MRA calves throughout the study, which suggests that these calves may have been experiencing an increase in systemic inflammation throughout the whole study. Increased fat intake from the MR (Amado et al., 2019) or increased osmolality stemming

from the increased intake of MR throughout the study may have led to gut permeability and an inflammatory response in the high-MRA calves (Wilms et al., 2019). As for SAA, the pattern of increased concentration in the first week, and its decrease throughout the study, has been demonstrated before (Tóthová et al., 2015; van Niekerk et al., 2021) and could potentially be due to transfer via colostrum (McDonald et al., 2001).

CONCLUSIONS

Our hypothesis that increased MR intake and higher starch in starter would lead to increased nutrient digestibilities, lower pH throughout the GIT, and increased systemic inflammatory markers was not observed. Although we found limited differences for increased starch intake (e.g., increased rumen pH magnitude, decreased rumen pH at dissection, and decreased haptoglobin in wk 8), the majority of differences came from increased starter intake in general. This suggests that with completely pelleted starters, the differences in starch and NDF do not elicit drastic changes in fermentation, subsequent end products, or any resulting inflammation in calves around weaning.

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